1. Explain the difference between Relative, Absolute and Mixed Cell Referencing. You have the following dataset:

Vehicle	Price
Car	500000
Bicycle	10500
Auto-Rickshaw	25000

Answer:

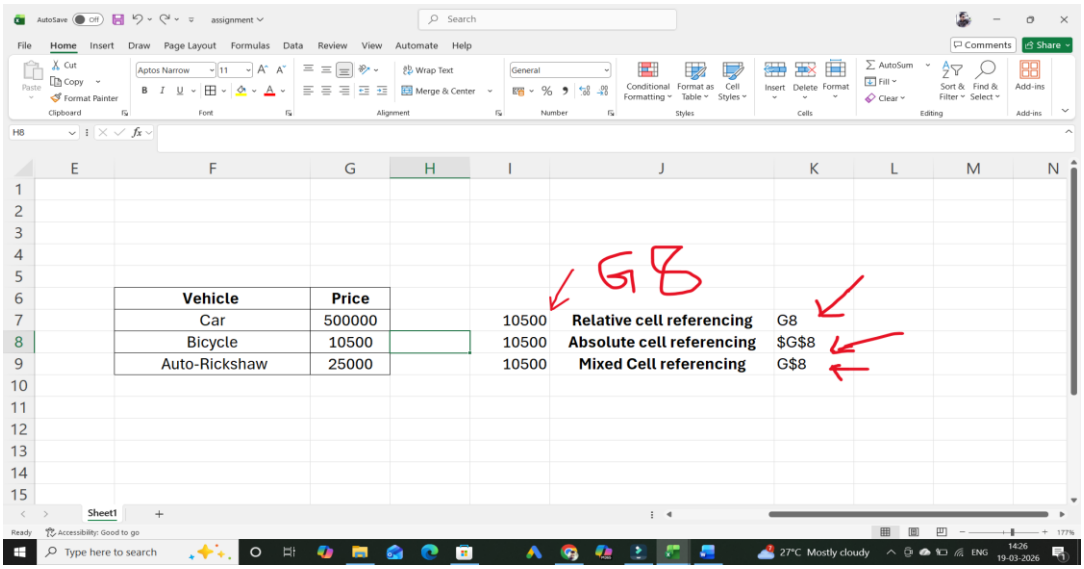
- *Relative Cell Referencing means when a cell reference changes automatically when the formula is copied to another cell.*
  For eg: =A1+B1

- *Absolute referencing is when a cell reference remains fixed, even when formula is copied and dragged down to other cells.*

  For eg: =$A$1+$B$2

- *Mixed cell referencing is when either row or column is fixed, while the other changes when formula is copied.*

  For eg: $A1/B$2



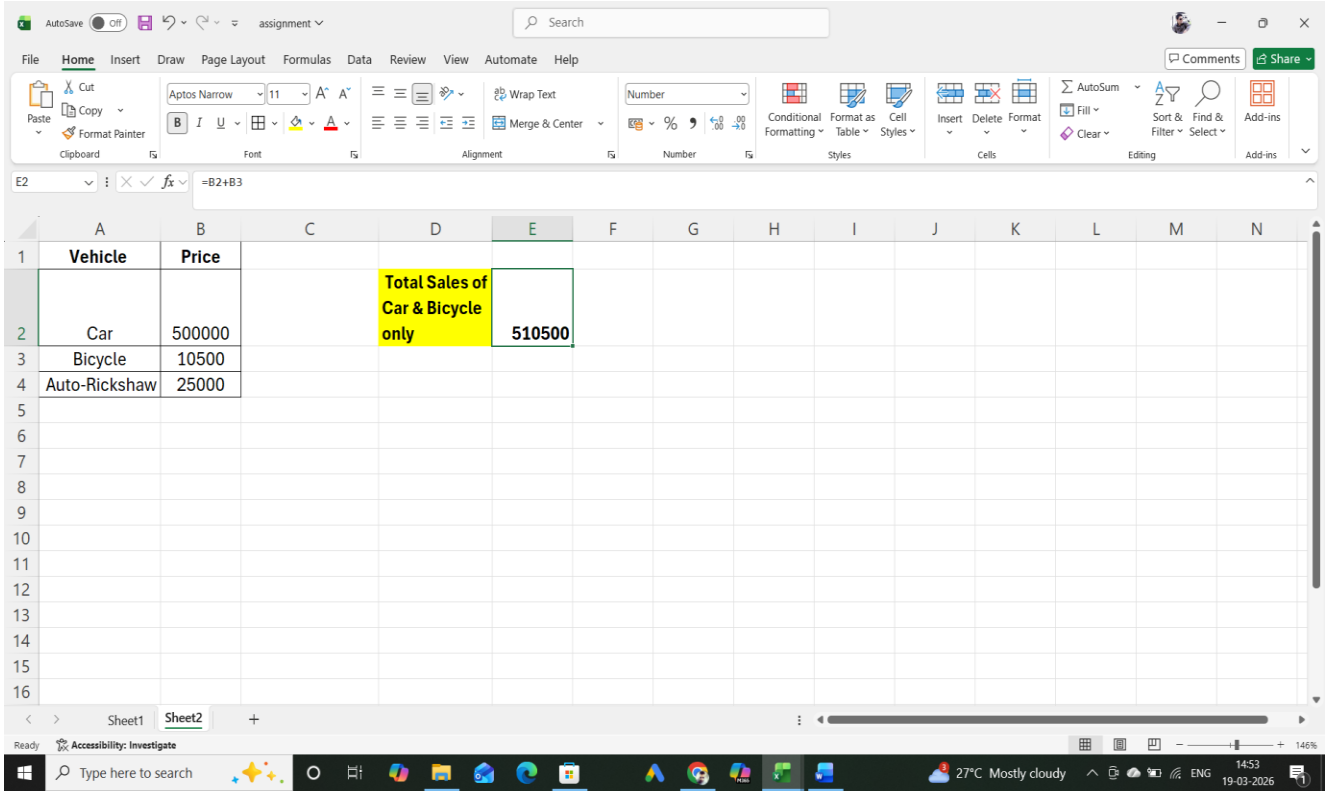
Question 2. Write a formula to calculate the total sales of Car and Bicycle only.

Vehicle	Price
Car	500000
Bicycle	10500
Auto-Rickshaw	25000

Answer:

Please also refer to the screenshot below:

- *Formula for Totals sales of Car & Bicycle is as under:*
- *=B2+B3*
- *Which will give result as **"510500".***

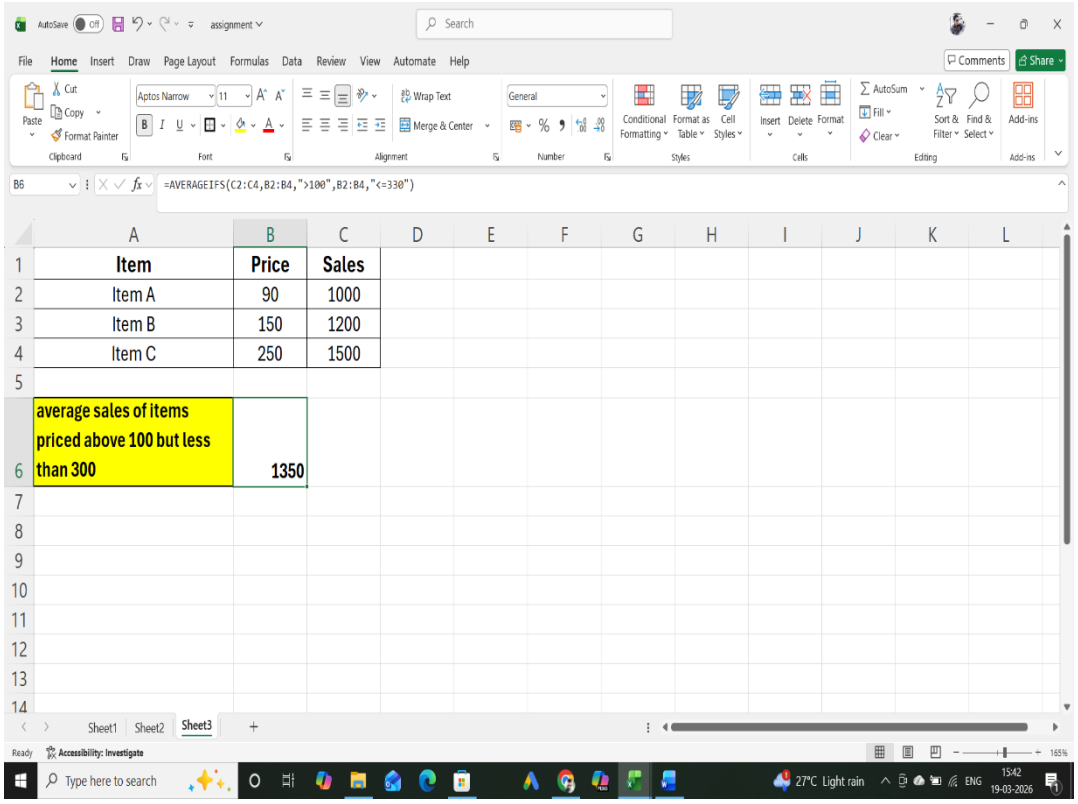


Question 3. Using the data below, write a formula to calculate the average sales of items priced above 100 but less than 300:

Item	Price	Sales
Item A	90	1000
Item B	150	1200
Item C	250	1500

Answer: Formula I have used for calculating the average sales of items priced above 100 but less than 300 is as under:

- *=AVERAGEIFS(C2:C4,B2:B4,">100",B2:B4,"<=330")*
- *Result is 1350*



Question 4. Count how many customer names are recorded.

Answer.

- *By using formula "=COUNTA(B2:B51)", we can count the customer names in the given dataset.*
- *For counting text values, we use above formula.*

Count	**50**

***Question 5**. Calculate the Total Sales for each row using a formula.*

Answer:

- *One more column was added in the add as "Column H" named as "Total Sales"*
- *The formula applied "=F2*E2" and formula copied and dragged down to the last row.*

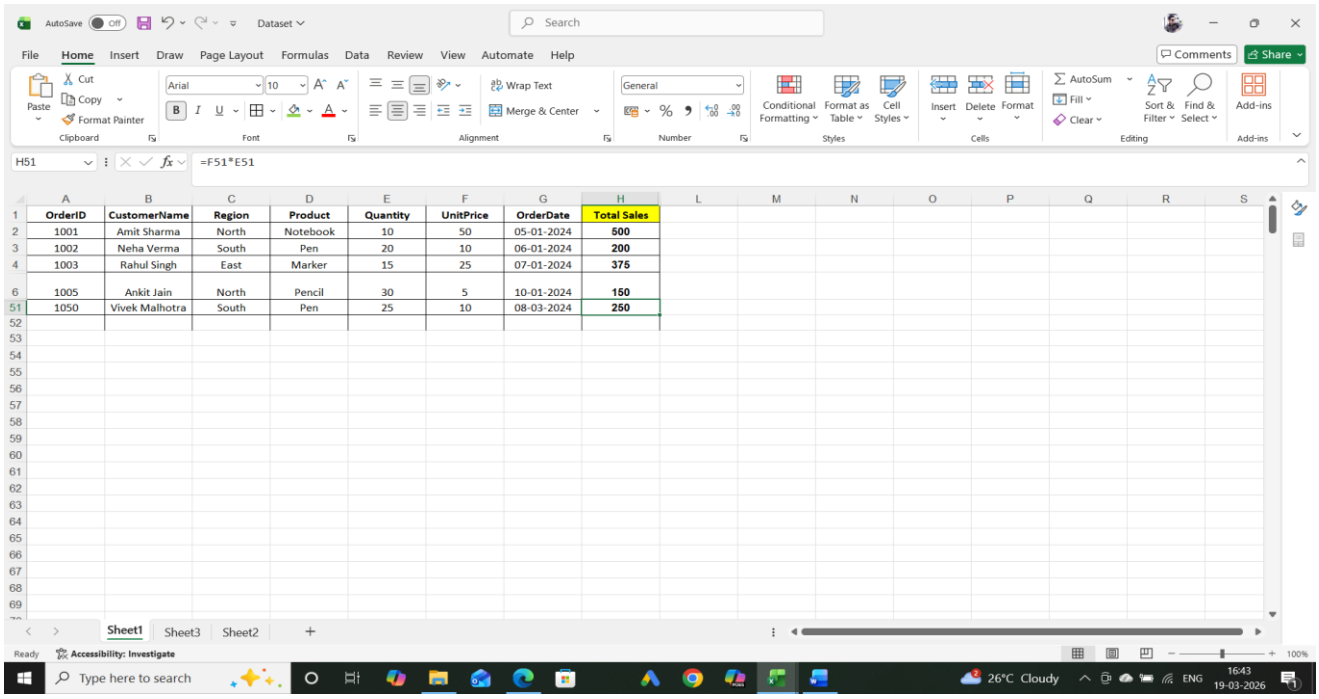
***Question 6**. Calculate the total sales of Notebooks in the North region only.*

Answer:

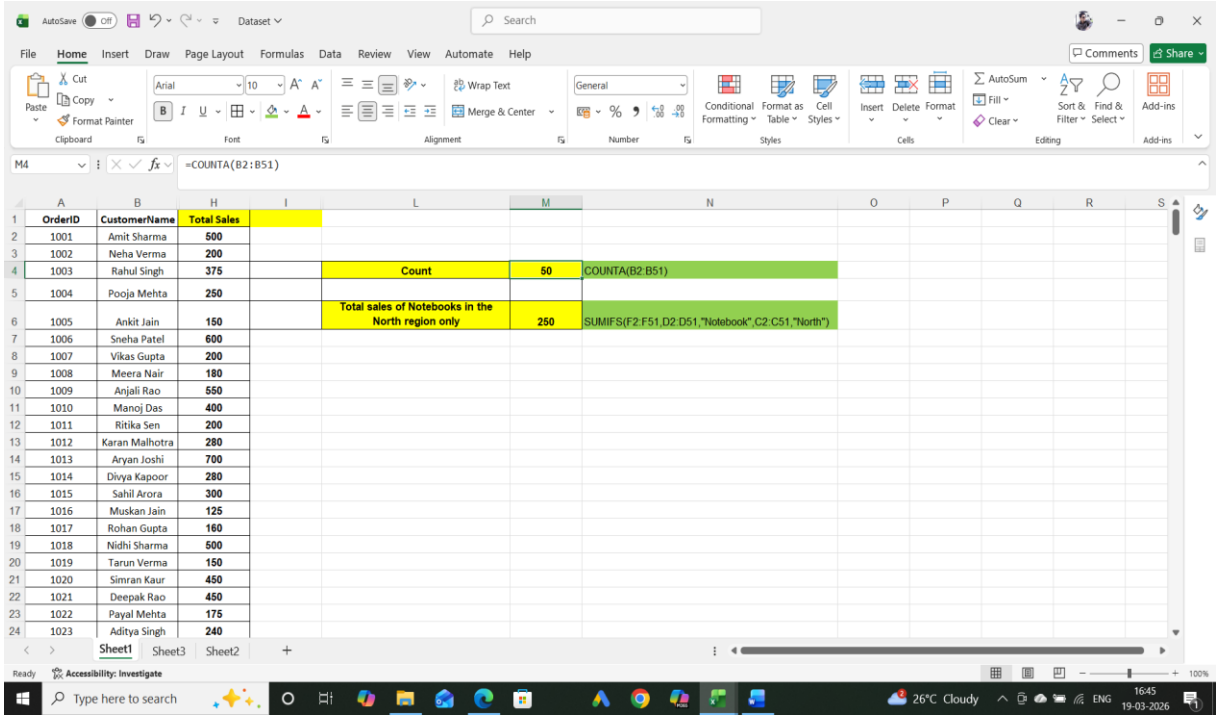
- *For calculating the sales with multiple conditions, we used "sumifs" formula*
- *=SUMIFS(F2:F51,D2:D51,"Notebook",C2:C51,"North")*

Total sales of Notebooks in the North region only	**250**

For explanation of question 4 - 6 , please refer the below screenshot:



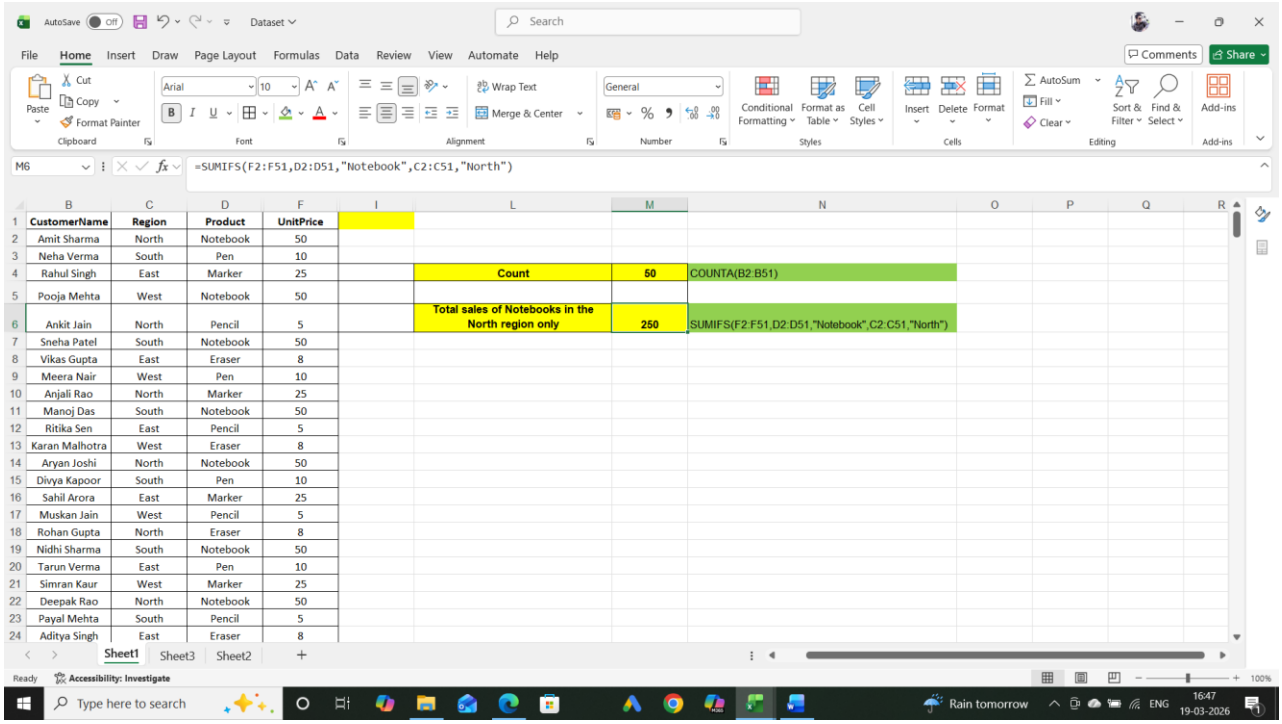
OrderID	CustomerName	Region	Product	Quantity	UnitPrice	OrderDate	Total Sales
1001	Amit Sharma	North	Notebook	10	50	05-01-2024	500
1002	Neha Verma	South	Pen	20	10	06-01-2024	200
1003	Rahul Singh	East	Marker	15	25	07-01-2024	375
1005	Ankit Jain	North	Pencil	30	5	10-01-2024	150
1050	Vivek Malhotra	South	Pen	25	10	08-03-2024	250



M4 =COUNTA(B2:B51)

OrderID	CustomerName	Total Sales
1001	Amit Sharma	500
1002	Neha Verma	200
1003	Rahul Singh	375
1004	Pooja Mehta	250
1005	Ankit Jain	150
1006	Sneha Patel	600
1007	Vikas Gupta	200
1008	Meera Nair	180
1009	Anjali Rao	550
1010	Manoj Das	400
1011	Ritika Sen	200
1012	Karan Malhotra	280
1013	Aryan Joshi	700
1014	Divya Kapoor	280
1015	Sahil Arora	300
1016	Muskan Jain	125
1017	Rohan Gupta	160
1018	Nidhi Sharma	500
1019	Tarun Verma	150
1020	Simran Kaur	450
1021	Deepak Rao	450
1022	Payal Mehta	175
1023	Aditya Singh	240

Count	50	COUNTA(B2:B51)
Total sales of Notebooks in the North region only	250	SUMIFS(F2:F51,D2:D51,"Notebook",C2:C51,"North")



M6 =SUMIFS(F2:F51,D2:D51,"Notebook",C2:C51,"North")

CustomerName	Region	Product	UnitPrice
Amit Sharma	North	Notebook	50
Neha Verma	South	Pen	10
Rahul Singh	East	Marker	25
Pooja Mehta	West	Notebook	50
Ankit Jain	North	Pencil	5
Sneha Patel	South	Notebook	50
Vikas Gupta	East	Eraser	8
Meera Nair	West	Pen	10
Anjali Rao	North	Marker	25
Manoj Das	South	Notebook	50
Ritika Sen	East	Pencil	5
Karan Malhotra	West	Eraser	8
Aryan Joshi	North	Notebook	50
Divya Kapoor	South	Pen	10
Sahil Arora	East	Marker	25
Muskan Jain	West	Pencil	5
Rohan Gupta	North	Eraser	8
Nidhi Sharma	South	Notebook	50
Tarun Verma	East	Pen	10
Simran Kaur	West	Marker	25
Deepak Rao	North	Notebook	50
Payal Mehta	South	Pencil	5
Aditya Singh	East	Eraser	8

Count	50	COUNTA(B2:B51)
Total sales of Notebooks in the North region only	250	SUMIFS(F2:F51,D2:D51,"Notebook",C2:C51,"North")

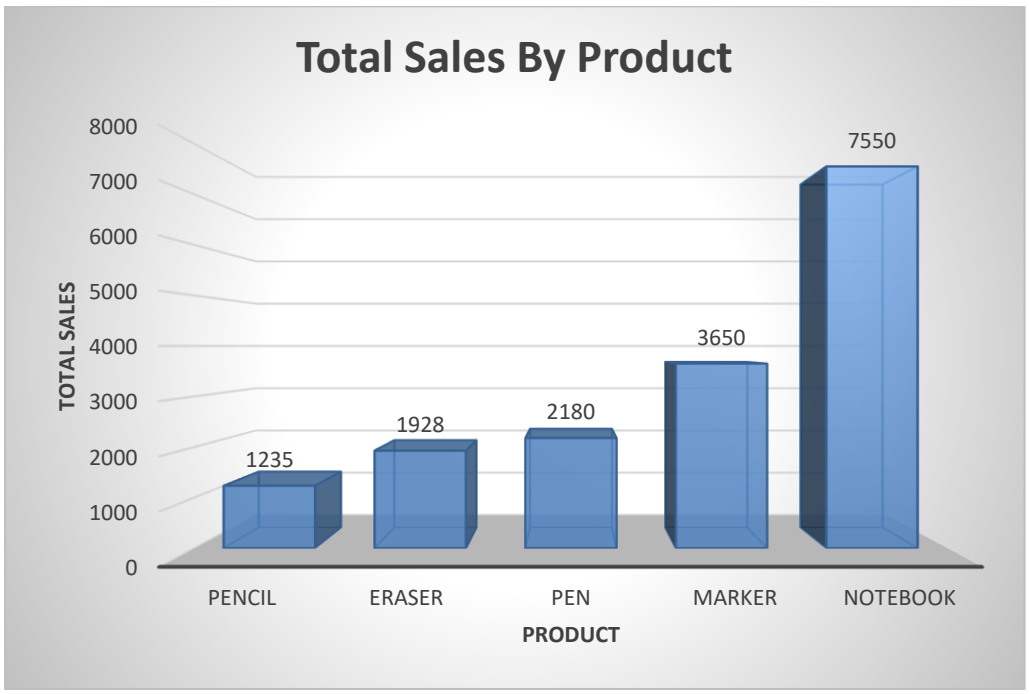
Question 7. *Create a* ***column chart*** *showing total sales by product.*

Answer:

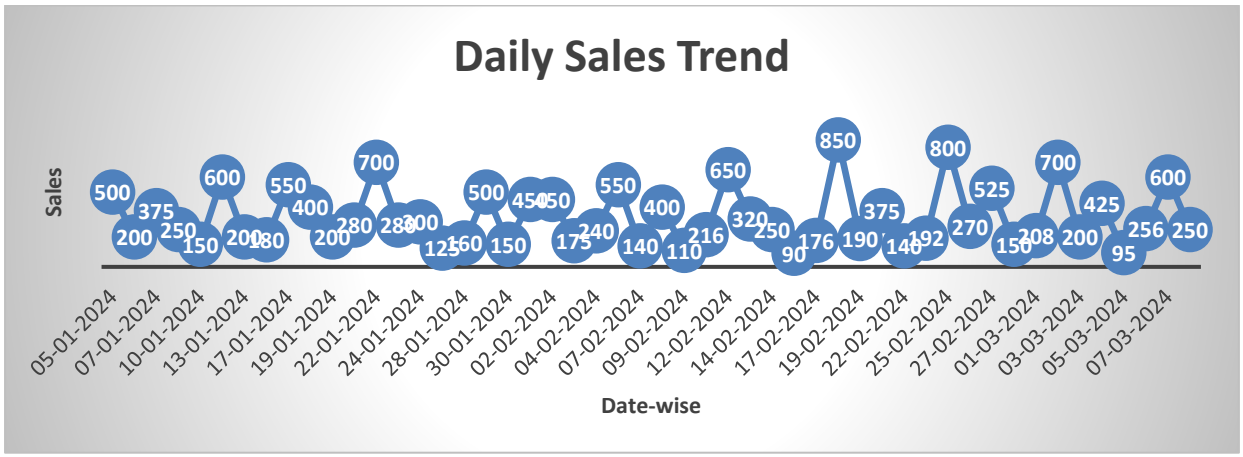
Answer:

- *Select the Product and TotalSales column*
- *Insert pivot table and copy product wise-> Total sales*
- *Insert chart type-> Column chart*

There were repetitive products in the Product tab, so pivot table used for accuracy and then insert chart type.



Question 8. Insert a line chart showing daily sales trend.

Answer:

- *Select the OrderDate and Totalsales columns*
- *Insert chart type-> line chart*
- *Add data labels and axis titles*